

- 7 Fig. 7.1 shows what is observed when a parallel beam of electrons, accelerated by a potential difference V , is incident on a fluorescent screen.

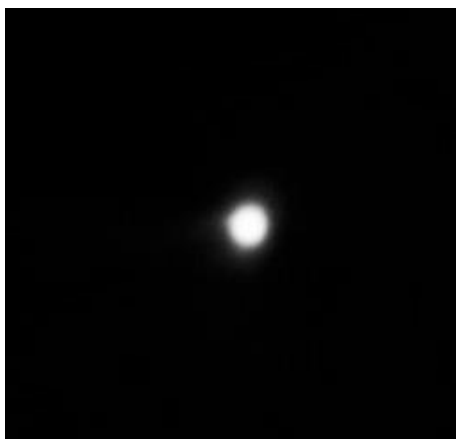


Fig. 7.1

A carbon film is then placed perpendicularly to the path of the electron beam as shown in Fig. 7.2.

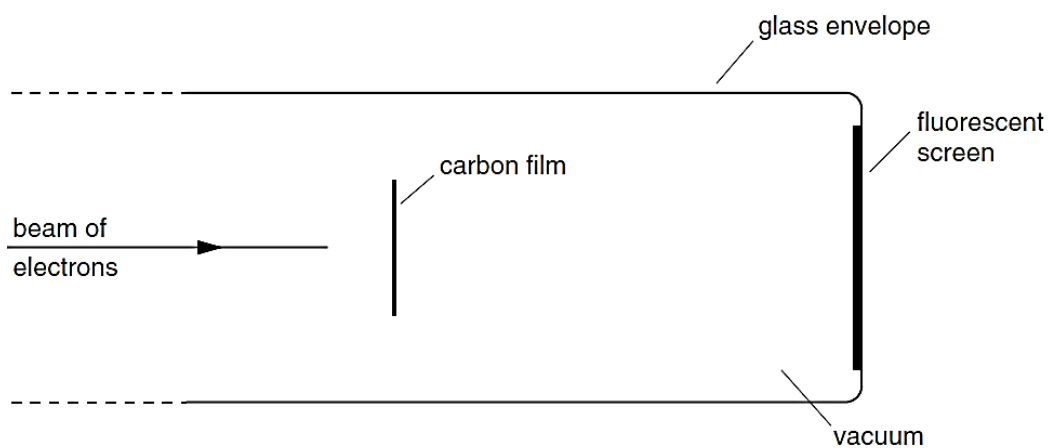


Fig. 7.2

The pattern observed on the screen is shown in Fig. 7.3.

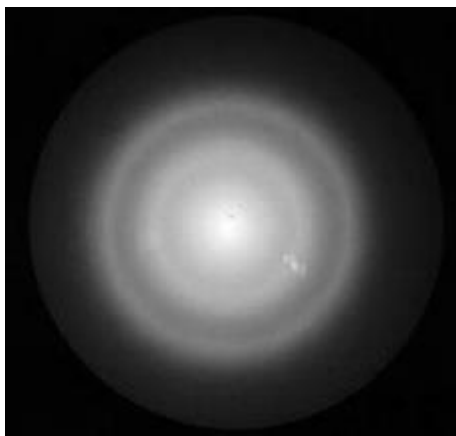


Fig. 7.3

- (a) Identify and explain two key features in Fig. 7.3 that provide evidence for the wave nature of the electrons.

Pattern spreading/occupy over a large area demonstrates diffraction of the beam. [B1] (don't accept diffract w/o explanation)

Bright rings (with alternating with dark regions between them) indicates regions of constructive and destructive interference. [B1] (don't accept interference w/o explanation)

Since both interference & diffraction are behaviours exhibited by waves, hence the electrons which are usually treated as particles, have wave-like behaviours. [B1] [3]

- (b) The electrons were accelerated through a potential difference of 2000 V.

- (i) Calculate the momentum of an electron after being accelerated through a potential difference of 2000 V.

$$E_k \text{ of electron} = \frac{p^2}{2m}$$

By conservation of energy,

Loss in electric potential energy = Gain in kinetic energy

$$2000 \times 1.60 \times 10^{-19} = \frac{p^2}{2(9.11 \times 10^{-31})} \quad [\text{M1}]$$

$$p = 2.41 \times 10^{-23} \text{ N s} \quad [\text{A1}]$$

momentum = N s [2]

- (ii) Hence, calculate the de Broglie wavelength of the electrons.

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{2.41 \times 10^{-23}} \quad [\text{C1}]$$

$$= 2.75 \times 10^{-11} \text{ m} \quad [\text{A1}]$$

wavelength = m [2]

(c) State and explain the changes, if any, that is observed in the pattern on the screen when the following changes are made separately.

(i) The potential difference used to accelerate the electrons is increased.

With a larger potential difference used, the kinetic energy/speed of the electrons will become larger and hence momentum becomes larger. With a larger momentum, the de Broglie wavelengths of the electrons become smaller. ^[M1] The rings will now have ^[A1] a smaller diameter/ closer together. ^[2]

(ii) The current of the electron beam is increased.

A greater rate of electrons is incident on the fluorescent screen. ^[M1]

This will cause the bright rings to appear brighter/greater contrast. ^[A1]

[2]

[Total: 11]

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Section B

Answer **one** question from this section in the spaces provided.

8 (a) Fig. 8.1 shows the variation of the gravitational potential ϕ with distance d from the surface